

Customer Churn Prediction Using Machine Learning and Clustering Techniques

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Abstract—Customer churn is a major concern for businesses as it directly affects revenue and growth. This project aims to predict customer churn using machine learning models and analyze customer behavior through clustering techniques. The dataset was preprocessed using encoding, imputation, feature selection, and feature scaling. Three supervised learning models—K-Nearest Neighbors, Decision Tree, and Neural Network—were trained and evaluated using accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC score. Additionally, K-Means clustering was applied as an unsupervised learning approach. The results show that Decision Tree achieved slightly better performance among the models, while Neural Network showed comparable results.

Index Terms—Customer Churn, Machine Learning, KNN, Decision Tree, Neural Network, K-Means, ROC Curve, AUC

I. INTRODUCTION

Customer churn refers to customers leaving a service or product over time. Predicting churn is important because retaining customers is more cost-effective than acquiring new ones. Machine learning techniques are widely used for churn prediction due to their ability to learn patterns from historical data [1].

In this project, different machine learning models are used to predict churn based on customer attributes. Both supervised and unsupervised learning techniques are applied to gain deeper insights into customer behavior.

II. DATASET DESCRIPTION

The dataset consists of customer demographic, financial, and behavioral features. The target variable is *Churn*, which is binary (0 = No, 1 = Yes).

Numerical Features: Age, Income, Credit Score, Number of Purchases, Membership Years **Categorical Features:** Gender, Marital Status, Device Used

The dataset contains 750 data points and 8 original features before encoding. After encoding, the number of features increased due to one-hot encoding.

III. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the dataset.

Boxplots were used to detect outliers in numerical features. Density plots were used to observe the distribution of numerical variables. Bar charts were used to analyze categorical feature distributions.

A correlation heatmap was also generated to identify relationships among features.

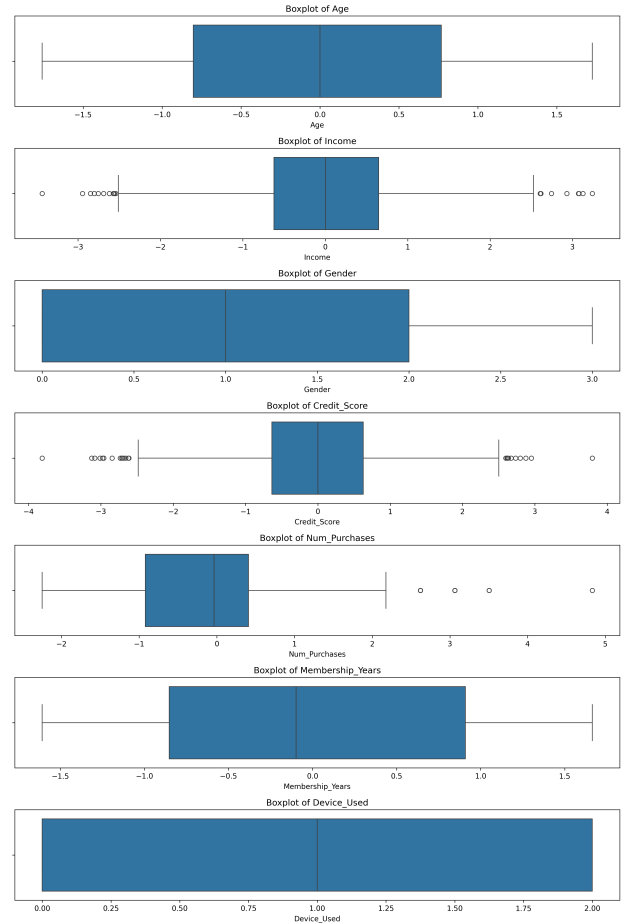


Fig. 1. Boxplot of numerical features showing outliers.

IV. DATASET PREPROCESSING

Data preprocessing was performed to improve model performance.

A. Handling Missing Values

Missing values were handled using SimpleImputer with mean strategy.

Density Plot of Numerical Features

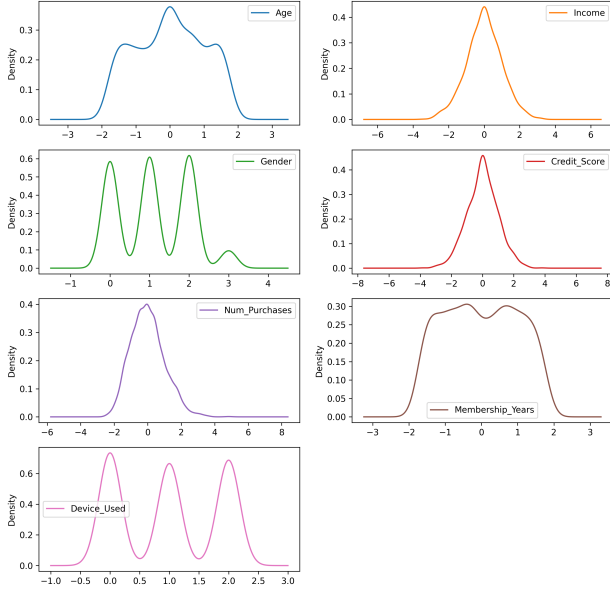


Fig. 2. Density plots of numerical features.

B. Encoding

Label encoding was applied to categorical variables, and one-hot encoding was applied to Marital Status.

C. Feature Selection

The feature *Marital_Status_Divorced* was removed based on correlation analysis.

D. Feature Scaling

StandardScaler was used to normalize numerical features. Scaling is important for distance-based models like KNN [2].

V. DATASET SPLITTING

The dataset was split into training and testing sets using a 70:30 ratio. Stratified sampling was used to maintain class balance.

VI. MODEL TRAINING

A. K-Nearest Neighbors

KNN is a distance-based algorithm that classifies data based on nearest neighbors [2].

B. Decision Tree

Decision Tree splits data into branches based on feature values and captures nonlinear relationships [3].

C. Neural Network

A Multi-Layer Perceptron (MLP) was used to capture complex patterns in data [4].

D. K-Means Clustering

K-Means clustering was used to group customers into clusters. The Elbow Method was used to determine the number of clusters [5].

VII. RESULTS AND EVALUATION

The models were evaluated using accuracy, precision, recall, and F1-score. The relatively low performance values (around 0.5) indicate that the dataset may not have strong separable patterns for churn prediction without further feature engineering.

Among the three models, Decision Tree showed slightly better performance in accuracy, precision, recall, and F1-score. However, the overall performance of all models was moderate, indicating that the dataset may require further feature engineering or model tuning.

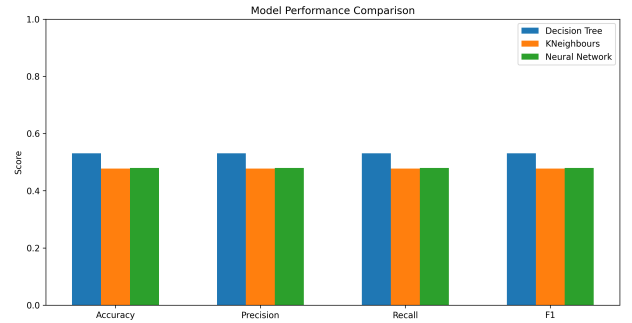


Fig. 3. Comparison of model performance.

A. Confusion Matrix

Confusion matrices were used to evaluate classification performance.

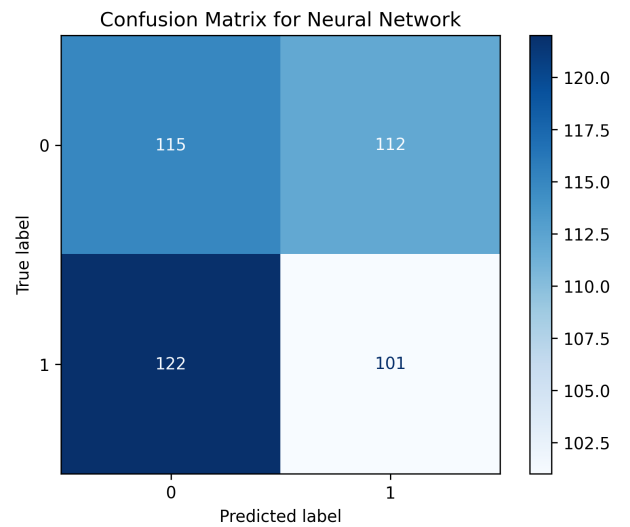


Fig. 4. Confusion matrix for Neural Network.

B. ROC Curve and AUC

ROC curves were plotted to evaluate classification performance. Decision Tree achieved the highest AUC score, while Neural Network showed comparable performance. [6].

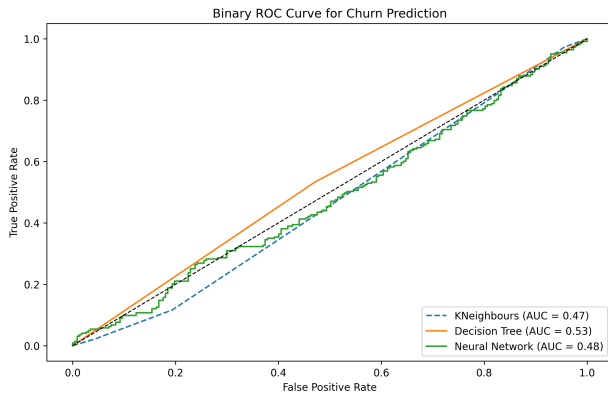


Fig. 5. ROC Curve comparison.

VIII. CLUSTERING ANALYSIS

K-Means clustering was applied to identify customer groups. The Elbow Method suggested 4 clusters. The silhouette score was used to evaluate clustering quality.

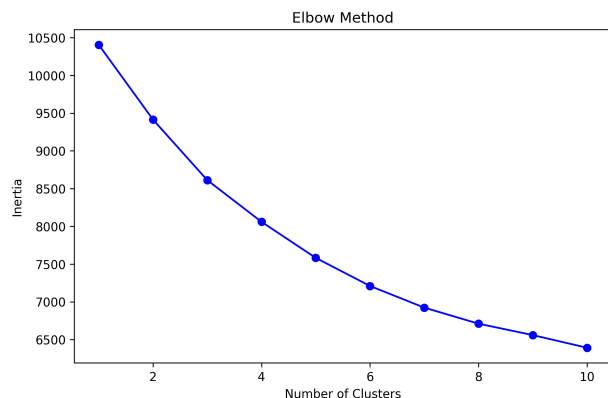


Fig. 6. Elbow Method for cluster selection.

IX. CONCLUSION

This project demonstrated the use of machine learning for customer churn prediction. Based on the evaluation results, Decision Tree achieved slightly better performance among the tested models, while Neural Network showed comparable performance. The results suggest that further preprocessing, hyperparameter tuning, or additional features may improve churn prediction. Decision Tree provided interpretable results, while KNN showed lower performance. K-Means clustering provided insights into customer segmentation.

ACKNOWLEDGMENT

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